Product Originality Detector using Blockchain

Dr. Benakappa S. M. [1], Rishabh Jayaram [2], Rishika [2], Rohith B. S [2], Sanketh S. T [2]
[1]Associate Professor, Department of Computer Science and Engineering, JNNCE, Shivamogga, Karnataka, India
[2]UG Students, Department of Computer Science and Engineering, JNNCE, Shivamogga, Karnataka, India

***Abstract*—In today's global market, counterfeit products pose a significant threat to brand reputation, consumer safety, and economic integrity. The project presents an Originality Product Identification System leveraging Blockchain technology and QR (Quick Response) codes to ensure the authenticity and traceability of products. The proposed system assigns each product a unique QR code, which is linked to a secure, tamper-proof record on a blockchain. At the time of manufacture, product data such as origin, manufacturer details, batch number, and timestamp are recorded on the blockchain. Consumers or stakeholders can scan the QR code using a smartphone application to instantly verify the product's originality and view its full history. The integration of blockchain guarantees data immutability and transparency, while QR codes provide a convenient and cost-effective interface for real-time verification. This system aims to combat counterfeiting, increase consumer trust, and improve supply chain transparency across various industries.**

Index Terms—*Blockchain, Anti-counterfeiting, Product authentication, Immutability, Smart Contracts.*

I. INTRODUCTION

Over the past few years, the spread of fake goods has become one of the burning issues on the international scale, negatively affecting consumer safety, company incomes, and brand image. With counterfeit products finding their way into industries as diverse as pharmaceuticals and electronics, as well as the luxury goods industry, even conventional verification processes have failed to offer any meaningful resolutions. As a way of combating this rising menace, researchers and technologists have looked to blockchain technology, in growing numbers, as an immutable, decentralized ledger that could store product authentication and supply chain transparency information.

Blockchain is a revolutionary way of allowing immutable data recording and traceability, where it is close to impossible to change the history of a product without the agreement of the whole network. That immutability has prompted an array of systems that combine QR codes, smart contracts, and cryptographic hashing to verify product authenticity. As an illustration, a [1] Blockchain-Based Management System (BCBM) may associate the barcode or QR code of each item with an unchangeable record on the blockchain, allowing customers to confirm the authenticity of their purchase by themselves.

This idea has been developed by a number of studies. The Blockchain and QR-based Product Analyzer (BQRPA) is efficient in cross-validating the authenticity of products with the aid of a distinct code attached to every item [2]. In a similar fashion, the combination of one-time printable QR codes and GPS (Global positioning system) tracking, most recently suggested in an anti-counterfeiting solution, offers manufacturers a highly secure system to guarantee brand authenticity, and safeguard consumer confidence [3]. This framework is also strengthened by using Ethereum blockchain and smart contracts, which provide product verification transparency and automation [4].

These innovations do not confine their existence to theoretical discourses. Field tests have shown that blockchain-based anti-counterfeiting systems are possible and accurate. An outstanding implementation is the use of encrypted QR codes and secure manufacturer-consumer interfaces to generate an immutable product journey, between production and point of sale [5]. In a different approach, QR codes are created through computational algorithms and cryptographic hashing, which offers quick and secure counterfeit identification using mobile applications [6].

Also, blockchain-based systems, in addition to detecting fake products also seek to enhance overall supply chain processes through traceability, reduced losses in the grey market, and responsible manufacturing [7]. The systems can decentralize control and bring transparency [8], which will allow all stakeholders, including manufacturers, distributors, retailers, and end-users to take an active role in enforcing product integrity.

To sum up, blockchain technology and product authentication systems (such as QR codes and smart contracts) have a synergistic effect and can be considered a new paradigm in the fight against counterfeit products. These technologies are still developing and being adopted, but they have a considerable potential to change the situation in the area of safe and clear worldwide trading.

II. LITERATURE SURVEY

Blockchain can improve product identification by creating open, safe and unalterable tracking. This survey treats the current researches to have a glimpse of the role they play in enhancing traceability in production systems.

In [9], the product tracking and counterfeiting are improved with the help of a blockchain and QR code-based solution. The QR code of each product is unique and points to the production data of the product on an unmodifiable blockchain. These codes can be verified by the stakeholders, enforced by smart contracts which enable the procedure with the least degree of error. A mobile application enables the consumer to find out

immediately whether it is genuine, which enhances trust and transparency. The system is removing the power imbalance between manufacturers, retailers, and consumers because it is built on Ethereum using smart contracts and role-based access. Nevertheless, set-up can be expensive, it is technical, QR codes are vulnerable, it is not scalable, and there are privacy concerns.

In [10], a decentralized system, built on Ethereum and utilizing smart contracts, is proposed in n to increase the visibility of the supply chain and fight counterfeiting. With the help of a Dapp (Decentralized app), manufacturers, suppliers, and customers cooperate to write and verify product information on the blockchain with QR codes. The platform is supported by technologies such as Ganache, Truffle, Web3.js, React and MetaMask. The product reliability, clear records, independent supply chain management, and consumer verification are its important advantages. Nonetheless, technical dependency, intensive computation expenses, a possible duplication of QR codes, and the necessity to be accepted by the vast majority of the market are among the challenges.

In [11], the system provides with single-use QR codes and Firebase to verify product authenticity, disabling codes after the first scan. It can be accessed through a React.js interface and provides real-time information, SHA (Secure Hash Algorithm) security, and convenience. Efficient as it is, it is not as secure as blockchain, requires network connection and could be subject to customer reluctance. Its capabilities may be improved with integration to blockchain and IoT.

In [12], a system based on blockchain with Hyperledger Fabric and Scratchable-QR codes is described where the products are authenticated and their ownership is traced through smart contracts and REST (Representational State Transfer) APIs. With the help of "Hyper Supply," stakeholders track the real-time supply chain. It increases security, transparency and consumer confidence, and has disadvantages such as being costly, relying on experts, low user acceptance, and scalability to small business.

The authors [13] have proposed a blockchain-based system which allows preventing counterfeiting and verifying products with the help of Ethereum smart contracts and special QR codes. It provides hash chains to secure it and KPI (Key Performance Indicator) tools to monitor it. Despite the confronted issues, such as low transaction throughput and usage, it improves data security, transparency, and brand protection, and can be prospective in regards to AI and IoT integration.

In [14], a POMS (Production and Operation Management Society) using blockchain that is implemented on the Ethereum platform tackles the problem of post-supply chain counterfeiting and RFID duplication. It attaches exclusive EPCs (Engineering Procurement Construction) to goods and applies smart contracts on RFID (Radio Frequency Identification tags), QR, or NFC (Near Field Communication) tags to ensure safe tracking. Although POMS some drawbacks, such as dependence on Ethereum, privacy, and brittle protocols, it has secure ownership transfer, consumer verification, and low operation costs.

In [15], the authors look at fraud in blockchain-based reputation systems and differentiate between fact-based and opinion-based fraud. Blockchain and cryptographic measures can minimise factual fraud, but not opinion-based fraud such as fake ratings on real transactions. Therefore, even with enhanced trust and security, extra verifications are necessary.

In [16], a product and customer data are secured in a private blockchain system with the signature of SHA-256 and barcode recovery allowing efficient verification with a QR code. It can provide trusted, cost-effective authentication and consumer confidence by allowing no alteration of records. Although it is scalable through cloud and key-based access, it has a product limit of 1,000 and requires QR readers, as well as depends on cloud performance.

In [17], the system described allows SMEs (Small and Medium Enterprises) to fight counterfeiting: it consists of a low-cost web-based system with hashing and QR codes, a MySQL database. Products are assigned an ID and a password, which are checked through the manufacture site, and the records are disabled once utilized. It is developed using PHP (Hypertext Preprocessor), HTML (Hyper Text Markup Language), and JavaScript and is quick, convenient, and cheap to verify, but it is centralized, which creates vulnerability and network connectivity issues.

The authors [18] have proposed a decentralized system based on Ethereum and smart contracts and web3.js to realize safe product authentication through QR codes. It is transparent and verifiable at low costs, since it is encrypted with ECDSA (Elliptic Curve Digital Signature Algorithm), has a PoW (Proof of Work) consensus mechanism, and ownership tracking modules. Although it is secure and decentralized, high-energy consumption, a limit on scalability, and the cost of entry are difficulties.

The authors [19], have proposed a blockchain-based system that involves QR codes to trace and authenticate the products safely. It has advantages like tamper evident records, real time verification and enhanced trust. Smart contracts guarantee the authorized individuals access the information. Nonetheless, external storage of the images is prone to tampering of the data, and QR code can be duplicated without further security implementation.

In [20], methodology entails permissioned blockchain, smart contracts to track, verify and to secure pharmaceutical supply chains. It provides immutability of data and verifies in real-time. Benefits are in terms of enhanced traceability, security, and curtailing of counterfeiting. Drawbacks include expensive set up and difficulty of integration.

In [21], a modified UTAUT2 (Unified theory of acceptance and use of technology) model applied in the research and analysed in SMART PLS (Partial Least Squares) allowed evaluating consumer preparedness towards blockchain-authenticated fashion. Its advantage is seen in the determination of the major drivers of adoption such as effort expectancy and social influence. But it is restricted by a thin and youthful sample and certain statistically inconsiderable variables.

The survey demonstrates that blockchain has the ability to improve product identification, bringing more transparency and preventing counterfeiting. The prospects are promising but such challenges as scalability and integration remain to be solved.

III. SUMMARY

Blockchain technology is presently seen as a safe and autonomous tool to help find original products and fight counterfeiting. Barcodes and RFID can be changed, but blockchain keeps tracking of each step of a product's journey from start to finish, so nothing is changed. Some researchers use platforms including Ethereum and Hyperledger and combine smart contracts and QR codes to ensure instant verification. Still, these systems do not work well for large volumes, with the IoT (Internet of Things) or when the costs are too high.

IV. CONCLUSION

Applications of blockchain in originality detection of products offers a transparent and safe way of authenticating products, which can help in the fight against counterfeit products. According to the literature, blockchain offers tamper-proof records because of its decentralized nature, which makes it a suitable solution in tracking the origin of products. The technology demonstrates excellent promise; however, the practical issues such as cost of implementation and integrating the system are subjects of on-going research and development.

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